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4.25

a) $X(j\omega) \rightarrow A(j\omega)e^{j\theta(j\omega)}$

$y(t) = x(t+1)$

$Y(j\omega) = A(j\omega)$

$\theta(j\omega) = -\omega$

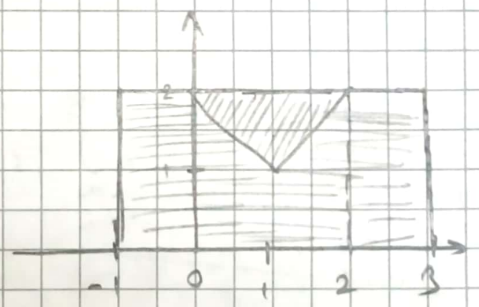
$y(j\omega) = X(j\omega)e^{j\omega}$

$X(j\omega) = Y(j\omega)e^{-j\omega}$ $\quad [A(j\omega)e^{j\theta(j\omega)}]$

b) $X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$ $\quad \{\omega=0\}$

$X(j0) = \int_{-\infty}^{\infty} x(t)e^{-j0t}dt$

$= \int_{-\infty}^{\infty} x(t)dt = 7$



Area triangle $= \frac{1}{2} \cdot 2 \cdot 1$

$= 1$

Area square $= 4 \cdot 2 = 8$

Area $= 8 - 1 = 7$

c) $x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}$ $\quad t=0$

$x(0) = \int_{-\infty}^{\infty} x(j\omega)d\omega = 2\pi x(0)$ $\quad \hookrightarrow 2$

$= 4\pi$

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d) $\int_{-\infty}^{\infty} X(j\omega)\frac{2\sin w}{w}e^{j2w}dw \rightarrow \int_{-\infty}^{\infty} x(j\omega)Y(j\omega)d\omega = 2\pi(x(t)*y(t))\Big|_{t=0}$

$y(t)\begin{cases}1, & |t|<T_1\\ 0, & |t|>T_1\end{cases}$ Fourier Transform $\frac{2\sin wT_1}{w} \Rightarrow \frac{2\sin w}{w}$

$T_1 = 1$

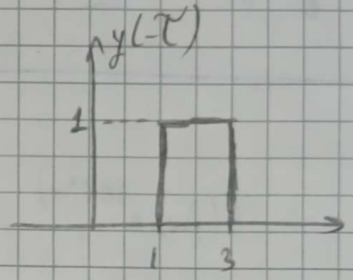
$y(t+t_0) \xrightarrow{F.T} Y(j\omega)e^{j\omega t_0}$

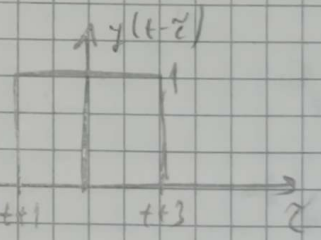
$y(t)\begin{cases}1, & -3<t<1\\ 0, & \text{otherwise}\end{cases} \xrightarrow{F.T} \frac{2\sin w}{w}e^{j2w}$

$x(t)*y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$

$= \int_{t+1}^{0} 2dt + \int_{0}^{t+3}(2-t)dt$

$= -2(t+1) + \left(2t - \frac{t^2}{2}\right)\Big|_0^{t+3}$





$\int_{-\infty}^{\infty} X(j\omega)Y(\omega)d\omega = 2\pi(x(t)*y(t))\Big|_{t=0}$

$\rightarrow 2+2(t+3)-\frac{(t+3)^2}{2}$ $\rightarrow 2\pi\cdot\left(\frac{3}{2}\right) = 3\pi$

$= 3/2$

e)

$x(t) = \begin{cases}2 & \text{for } -1<t<0\\ 2-t & 0<t<1\\ t & 1<t<2\\ 2 & 2<t<3\\ 0 & \text{otherwise}\end{cases}$

$\int_{-\infty}^{\infty}|X(j\omega)|^2 d\omega = 2\pi\int_{-\infty}^{\infty}x^2(t)dt$

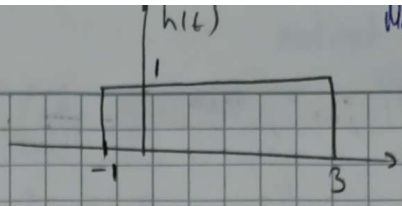
$\int_{-\infty}^{\infty}x^2(t)dt = \int_{-1}^{0}\underbrace{x^2(t)}_{(2)^2}dt + \int_{0}^{1}\underbrace{x^2(t)}_{(2-t)^2}dt + \int_{1}^{2}\underbrace{x^2(t)}_{t^2}dt + \int_{2}^{3}\underbrace{x^2(t)}_{(2)^2}dt$

$= 4t\Big|_{-1}^{0} + 4t\Big|_0^1 - 4\left(\frac{t^2}{2}\right)\Big|_0^1 + 4\frac{t^3}{3}\Big|_1^2 + \frac{t^3}{3}\Big|_2^3 + 4t\Big|_2^3$

$= 13,67 \cdot 2\pi = 27,33\pi$

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4.26 (a)



i) $x(t) = te^{-2t}u(t)$, $h(t) = e^{-4t}u(t)$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

$$= \int_{0}^{\infty} te^{-t(2+j\omega)}\, dt \quad (\text{since, } u(t) = 0 \text{ for } t<0)$$

$$X(j\omega) = t\int_{0}^{\infty} e^{-t(2+j\omega)}\, dt - \int_{0}^{\infty}(1)\left(\int_{0}^{\infty} e^{-t(2+j\omega)}\, dt\right) dt$$

$$= \left[\frac{te^{-t(2+j\omega)}}{-(2+j\omega)}\right]_{0}^{\infty} - \int_{0}^{\infty} \frac{e^{-t(2+j\omega)}}{-(2+j\omega)}\, dt$$

$$= \frac{1}{(2+j\omega)^2}$$

$$H(j\omega) = \int_{-\infty}^{\infty} h(t)\, e^{-j\omega t}\, dt$$

$$\int_{0}^{\infty} e^{-(4+j\omega)t}\, dt = \frac{1}{4+j\omega}$$

$$Y(j\omega) = H(j\omega) \times X(j\omega)$$

$$= \left[\frac{1}{(2+j\omega)^2}\right]\left[\frac{1}{4+j\omega}\right]$$

$$= \frac{1}{(2+j\omega)^2(4+j\omega)}$$

$$\frac{A}{2+j\omega} + \frac{B}{(2+j\omega)^2} + \frac{C}{(4+j\omega)}$$

$$1 = A(2+j\omega)(4+j\omega) + B(4+j\omega) + C(2+j\omega)^2$$

$j\omega = -2 \quad B = \frac{1}{2}$

$j\omega = -4 \quad C = \frac{1}{4}$

$$\Rightarrow \frac{1}{4}\left(-e^{-2t} + 2te^{-2t} + e^{-4t}\right)u(t)$$

ii) $x(t) = te^{-2t}$, $h(t) = te^{-4t}$

$$\rightarrow X(j\omega) = \frac{1}{(2+j\omega)^2}$$

$$\rightarrow H(j\omega) = \frac{-1}{(4+j\omega)^2}$$

$$Y(j\omega) = X(j\omega)\, H(j\omega) = \frac{1}{(2+j\omega)^2(4+j\omega)^2}$$

$$Y(j\omega) = \frac{-\frac{1}{4}}{2+j\omega} + \frac{\frac{1}{4}}{(2+j\omega)^2} + \frac{1/4}{4+j\omega} + \frac{1}{(4+j\omega)^2} \Rightarrow y(t) = -\frac{1}{4}e^{-2t}u(t) + \frac{1}{4}te^{-2t}u(t) + \frac{1}{4}e^{-4t}u(t)$$

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(a) (ii) $X(j\omega) = 1/(1+j\omega)$, $H(j\omega) = \frac{1}{1+j\omega}$ $\Rightarrow$ $Y(j\omega) = X(j\omega)H(j\omega) = 1/(1-j\omega)(1+j\omega)$

$$Y(j\omega) = \frac{1/2}{1+j\omega} + \frac{1/2}{1-j\omega} = \frac{1}{2}\left[e^{-t}u(t) + e^{t}u(-t)\right]$$

(b) $h(t) = u(t+1) - u(t-3) \Rightarrow y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau = \int_{2}^{t+1} e^{-(\tau-2)}\,d\tau = 1 - e^{-(t-1)}$

$$Y(j\omega) = \int_{1}^{5}\left[1-e^{-(t-1)}\right]e^{-j\omega t}\,dt + \int_{1}^{5}\left[e^{-(t-5)} - e^{-(t-1)}\right]e^{-j\omega t}\,dt$$

$$+ \int_{t-3}^{t+1} e^{-(\tau-2)}\,d\tau = -e^{2}\left[e^{-(t+1)} - e^{-(t-3)}\right]$$

$$= \left(\frac{2e^{-3j\omega}}{1+j\omega}\right)\left(\frac{\sin 2\omega}{\omega}\right)$$

6.29) a)

(i) $Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega}) \Rightarrow X(e^{j\omega}) = 1/(1-\frac{3}{4}e^{-j\omega}),\ H(e^{j\omega}) = 1/(1-\frac{1}{2}e^{-j\omega})$

$$= \frac{-2}{1-\frac{1}{2}e^{-j\omega}} + \frac{3}{1-\frac{1}{4}e^{-j\omega}} \Rightarrow y[n] = 3\left(\frac{3}{4}\right)^{n}u[n] - 2\left(\frac{1}{2}\right)^{n}u[n]$$

(ii) $X(e^{j\omega}) = 1/(1-\frac{1}{4}e^{-j\omega})^2,\ H(e^{j\omega}) = 2/(1-\frac{1}{2}e^{-j\omega})$

$$Y(e^{j\omega}) = \frac{4}{1-\frac{1}{2}e^{-j\omega}} - \frac{2}{1-\frac{1}{4}e^{-j\omega}} - \frac{1}{(1-\frac{1}{4}e^{-j\omega})^2} \Rightarrow y[n] = 4\left(\frac{1}{2}\right)^{n}u[n] - 2\left(\frac{1}{4}\right)^{n}u[n]$$

$$-(n+1)\left(\frac{1}{4}\right)^{n}u[n]$$

(iii) $X(e^{j\omega}) = 2\pi\sum_{k=-\infty}^{\infty}\delta[\omega-(2k+1)\pi],\ H(e^{j\omega}) = 1/(1-\frac{1}{3}e^{-j\omega})$

$$Y(e^{j\omega}) = \frac{2}{3}\left(\pi\sum_{k=-\infty}^{\infty}\delta(\omega-\pi-2\pi k)\right) \Rightarrow y[n] = \frac{2}{3}(e^{j\pi})^{n} = \frac{2}{3}(-1)^{n}$$

(b) (i) $X(e^{j\omega}) = 1/(1-\frac{1}{2}e^{-j\omega}),\ H(e^{j\omega}) = \left(\frac{1/2}{1-\frac{j}{2}e^{-j\omega}}\right) + \frac{1}{2}\cdot\frac{1}{1-\frac{1}{2}e^{-j\pi/2}e^{-j\omega}}$

$$Y(e^{j\omega}) = \frac{1}{2}\left(\frac{1}{1-\frac{1}{2}e^{-j\omega}}\right) + \frac{j}{2(j-1)}\left(\frac{1}{1-\frac{j}{2}e^{-j\omega}}\right) + \frac{j}{2(j+1)}\left(\frac{1}{1+\frac{j}{2}e^{-j\omega}}\right)$$

$$\Rightarrow y[n] = \frac{1}{2}\left(\frac{1}{2}\right)^{n}u[n] + \frac{j}{2(j-1)}\left(\frac{j}{2}\right)^{n} + \frac{j}{2(j+1)}\left(\frac{-j}{2}\right)^{n}$$

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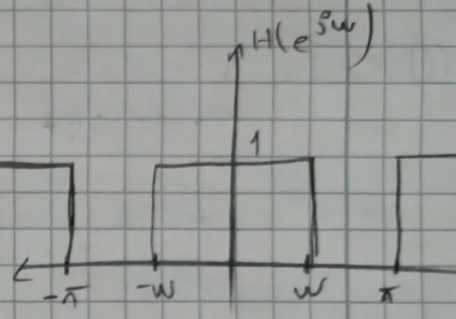
i) $X(e^{j\omega}) = \int\left(\omega - \frac{\pi}{2} - 2\pi l\right) + \int\left(\omega + \frac{\pi}{2} - 2\pi l\right)$

$Y(e^{j\omega}) = \frac{4}{3}\pi \sum_{l=-\infty}^{\infty} \int\left(\omega - \frac{\pi}{2} - 2\pi l\right) + \int\left(\omega + \pi/2 - 2\pi l\right)$

$\Rightarrow y[n] = \frac{4}{3}\cos\left(\frac{\pi n}{2}\right) u[n]$

5.30)

a) $H(e^{j\omega}) = \begin{cases} 1, & 0 \le |\omega| \le W \\ 0, & W < |\omega| \le \pi \end{cases} \Rightarrow$



b) i) $h[n] = \frac{\sin\left(\frac{\pi n}{6}\right)}{\pi n} \Rightarrow H(e^{j\omega}) = \begin{cases} 1, & 0 \le |\omega| \le \pi/6 \\ 0, & \pi/6 \le |\omega| \le \pi \end{cases}$ cut of freq $\Rightarrow \pi/6$

$y[n] = (1)\sin(\pi n/8) - (0)\, 2\cos(\pi n/4) = \frac{\sin \pi n}{8}$ //

ii) $H(e^{j\omega}) = \begin{cases} 1, & 0 \le |\omega| \le \pi/6 \\ 0, & \pi/6 \le |\omega| \le \pi \end{cases}$, $H_2(e^{j\omega}) = \begin{cases} 1, & 0 \le |\omega| \le \pi/2 \\ 0, & \pi/2 \le |\omega| \le \pi \end{cases}$

$y_1[n] = (2)\frac{\sin \frac{\pi n}{8}}{}$, $y_2[n] = (-2)\cos\left(\frac{\pi}{4} n\right)$

$y[n] = y_1[n] + y_2[n] = 2\sin\frac{\pi n}{8} - 2\cos\frac{\pi n}{4}$

iii) $H_1(e^{j\omega}) = \begin{cases} 1, & 0 \le |\omega| \le \pi/6 \\ 0, & \pi/6 \le |\omega| \le \pi \end{cases}$ $H_2(e^{j\omega}) = \begin{cases} 1, & 0 \le |\omega| \le \pi/3 \\ 0, & \pi/3 \le |\omega| \le \pi \end{cases}$

$H(e^{j\omega}) = H_1(e^{j\omega}) * H_2(e^{j\omega}) = \frac{1}{6} \cdot \frac{1}{2\pi}(\omega - \pi/6)$ for $\frac{\pi}{6} < \omega \le \pi/2$

$y_1[n] = \frac{1}{6}\sin\frac{\pi n}{8}$ $H(e^{j\omega}) = \frac{1}{8} \rightarrow y_2[n] = \frac{1}{8}\, 2\cos\frac{\pi n}{4}$

$y[n] = y_1[n] + y_2[n] = \frac{1}{6}\sin\frac{\pi n}{8} - \frac{1}{4}\cos\frac{\pi n}{4}$

iv) $H(e^{j\omega}) = \begin{cases} 1, & 0 \le |\omega| \le \pi/6 \\ 0, & \pi/6 \le |\omega| \le \pi \end{cases} \Rightarrow y_1[n] = (0)\sin\frac{\pi n}{8}$

$y_2[n] = (-2)\cos\left(\frac{\pi n}{2}\right)\left(\frac{1}{2\pi}\right) = -\sin\frac{\pi n}{4}$

$= y[n]$

Q) i) $x[n] = \begin{cases} 1, & -2 \le n \le 2 \\ 0, & 3 \le n \le 5 \end{cases} \Rightarrow X(e^{j\omega}) = \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi k}{N}\right) = \pi a_k \delta\left(\omega - \frac{\pi k}{4}\right)$

$$H(e^{j\omega}) = \begin{cases} 1, & 0 \le |\omega| \le \pi/3 \\ 0, & \pi/3 < |\omega| \le \pi \end{cases}$$

$$Y(e^{j\omega}) = 2\pi a_0 \delta(\omega) + 2\pi a_1 \delta(\omega - \pi/4) + 2\pi a_{-1} \delta(\omega + \pi/4) = a_0 + a_1 e^{j\pi/4} + a_{-1} e^{-j\pi/4}$$

ii) $x[n] = \begin{cases} 1, & n = 0 \\ 2, & 1 \le n \le 7 \end{cases} \Rightarrow X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta\left(\omega - \frac{\pi k}{4}\right)$

$$Y(e^{j\omega}) = 2\pi a_0 \delta(\omega) + 2\pi a_1 \delta(\omega - \pi/4) + 2\pi a_{-1} \delta(\omega + \pi/4) = a_0 + a_1 e^{j\pi/4} + a_{-1} e^{-j\pi/4}$$

$$\Rightarrow y[n] = \frac{1}{8} + \frac{1}{4} \cos\frac{\pi n}{4}$$

iii) $x[n] = \begin{cases} -1, & n = -1, 1 \\ 1, & n = -2, 0, 2 \\ 0, & n = 3, 4, 5 \end{cases} \Rightarrow X(e^{j\omega}) = \frac{2\pi}{N} \sum_{-\infty}^{\infty} \delta\left(\omega - \frac{2\pi k}{N}\right) = \sum_{-\infty}^{\infty} 2\pi \delta\left(\omega - \frac{\pi k}{4}\right)$

$$Y(e^{j\omega}) = 2\pi a_0 \delta(\omega) + 2\pi a_1 \delta(\omega - \pi/4) + 2\pi a_{-1} \delta(\omega + \pi/4)$$

$$y[n] = a_0 + a_1 e^{j\pi/4} + a_{-1} e^{-j\pi/4} = \frac{1}{8} + \frac{1}{4} \cos\frac{\pi n}{4} - \frac{1}{2\sqrt{2}} \cos\frac{\pi n}{4}$$

iv) $x[n] = \delta[n+1] + \delta[n-1]$

$$h[n] = \frac{\sin\frac{\pi n}{3}}{\pi n}, \quad y[n] = x[n] * h[n] = \frac{\sin\left[\frac{\pi(n+1)}{3}\right]}{\pi(n+1)} + \frac{\sin\left[\frac{\pi(n-1)}{3}\right]}{\pi(n-1)}$$

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